



AI-Powered Talent Scouting Agent

Architecture & Scoring Design by Gouripriya



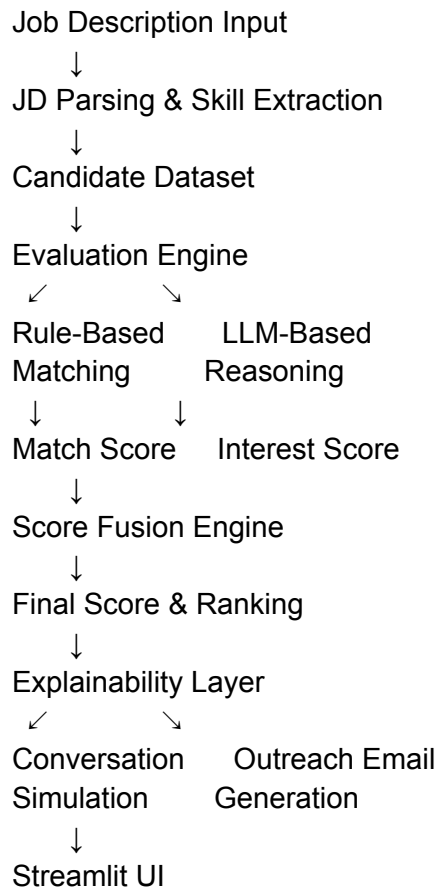
1. System Overview

The **AI-Powered Talent Scouting Agent** is designed to automate candidate discovery, evaluation, and engagement.

It processes a job description, evaluates candidates across **technical fit and behavioral intent**, and produces a **ranked shortlist with explainability and outreach generation**.



2. System Architecture



3. Core Components

1. JD Parsing & Skill Extraction

- Extracts key skills from the job description using LLM
- Normalizes skills for matching

2. Candidate Evaluation Engine

- Combines:
 - Rule-based logic (deterministic)
 - LLM reasoning (adaptive)

3. Scoring System

Produces:

- Match Score (technical)
- Interest Score (behavioral)
- Final Score (combined)

4. Explainability Layer

Provides:

- Technical reasoning
- Interest reasoning
- Final decision explanation

5. Interaction Layer

- Simulated recruiter-candidate conversation
- Personalized outreach email

4. Scoring Logic

♦ Match Score (Technical Fit)

- Based on **skill overlap**
- Formula:

Match Score = (Matched Skills / Total Required Skills) × 100

- Deterministic and explainable

Interest Score (Behavioral Fit)

- Derived from AI-generated conversation
- Evaluates:
 - engagement
 - alignment
 - willingness

Final Score (Fusion Model)

Final Score = 0.6 × Match Score + 0.4 × Interest Score

- Weighted to prioritize technical fit while considering intent

Candidate Classification

Category	Condition
Enthusiastic	Interest Score > 50
Neutral	40 – 50
Rejected	< 40

5. Robustness & Reliability

- Fallback scoring when LLM parsing fails
- Hybrid system ensures:
 - stability
 - consistency
 - reduced hallucination risk

6. Key Innovations

- Hybrid scoring (rule + AI)
- Explainable decision-making
- Simulated candidate engagement
- End-to-end recruitment pipeline

7. Output

The system delivers:

- Ranked candidate shortlist
- Multi-dimensional scoring
- Transparent explanations
- Simulated conversations
- Ready-to-send outreach email